Drive systems & mechanisms

National 5 · Mechanisms & structures · Answers

Answers and marking guidance

Award one mark per correct, developed point unless stated. Other valid answers should be credited.

2 Describe one way to reduce wear.

- Lubricate with oil or grease to reduce friction
- Use bearings
- Use harder or self-lubricating materials

3 Quick check

- a) Reciprocating motion
- b) 3:1
- c) 6500 rev/min
- d) No different

4 Key terms

- 1 – A
- 2 – D
- 3 – C
- 4 – E
- 5 – F
- 6 – B

5 Calculate the velocity ratio required.

- VR = input speed ÷ output speed = 1200 ÷ 25
- VR = 48 (48:1)

6 Calculate the velocity ratio and the output speed.

- VR = driven ÷ driver = 60 ÷ 20 = 3
- Output speed = 1500 ÷ 3
- = 500 revs min^{-1}

7 Calculate the output speed.

- VR1 = 45 ÷ 15 = 3; VR2 = 50 ÷ 10 = 5
- Total VR = 15
- Output = 3000 ÷ 15 = 200 revs min^{-1}